Complete Derivation of the κ Bound: The Combinatorial Structure

Generated from PRZZ Framework

1 Overview: The Main Result

We compute the proportion κ of Riemann zeta zeros on the critical line via:

$$\boxed{\kappa \geq 1 - \frac{\log c}{R}} \tag{1}$$

where $R = 1.3036$ is the shift parameter and c is the main-term constant from the mollified mean square.

Key Result: By optimizing the mollifier polynomials P_2 and P_3, we achieve:

$$\begin{aligned} \text{Baseline (PRZZ):} \quad & \kappa = 0.4173, \quad c = 2.137 \\ \text{Optimized } (\alpha = 61): \quad & \kappa = 0.4493, \quad c = 2.050 \quad \mathbf{(+7.68\%)} \end{aligned}$$

2 The Mollifier Structure

2.1 Three-Piece Mollifier ($K = 3$)

The mollifier has the form:

$$\psi(s) = \sum_{n \leq N} \frac{a_n}{n^s} = \sum_{\ell=1}^{3} \psi_\ell(s) \tag{2}$$

where $N = T^\theta$ with $\theta = \frac{4}{7}$, and each piece is:

$$\psi_1(s) = \sum_{n \leq N} \frac{\mu(n)}{n^s} P_1\left(\frac{\log(N/n)}{\log N}\right) \tag{3}$$

$$\psi_2(s) = \frac{1}{\log N} \sum_{n \leq N} \frac{(\mu \star \Lambda)(n)}{n^s} P_2\left(\frac{\log(N/n)}{\log N}\right) \tag{4}$$

$$\psi_3(s) = \frac{1}{(\log N)^2} \sum_{n \leq N} \frac{(\mu \star \Lambda \star \Lambda)(n)}{n^s} P_3\left(\frac{\log(N/n)}{\log N}\right) \tag{5}$$

2.2 Polynomial Definitions

2.2.1 Baseline PRZZ Polynomials

$$\begin{aligned} P_1(x) &= x + 0.261076\,x(1-x) - 1.071007\,x(1-x)^2 \\ &\quad - 0.236840\,x(1-x)^3 + 0.260233\,x(1-x)^4 \\ P_2(x) &= 1.048274\,x + 1.319912\,x^2 - 0.940058\,x^3 \\ P_3(x) &= 0.522811\,x - 0.686510\,x^2 - 0.049923\,x^3 \end{aligned}$$

2.2.2 Optimized Polynomials ($\alpha = 61$)

$$\begin{aligned} P_1(x) &= \text{(unchanged)} \\ P_2(x) &= 1.229608\,x + 0.688579\,x^2 - 0.584760\,x^3 \\ P_3(x) &= -0.645119\,x - 1.817008\,x^2 - 0.102188\,x^3 \end{aligned}$$

2.3 The Q Polynomial

The Q polynomial arises from residue calculus and appears in all integral kernels:

$$Q(x) = \sum_{j=0}^{5} q_j x^j \tag{6}$$

with coefficients in monomial basis:

$$Q = [1.000, -0.638, -0.631, -1.286, 2.561, -1.024]$$

3 The Difference Quotient Identity

Proposition 1 (PRZZ Lines 1508-1511)**.** *For $N = T^\theta$ and complex α, β with* $\mathrm{Re}(\alpha), \mathrm{Re}(\beta) > -1$*:*

$$\frac{N^{\alpha x+\beta y} - T^{-\alpha-\beta} N^{-\beta x-\alpha y}}{\alpha+\beta} = N^{\alpha x+\beta y} \log(N^{x+y}T) \int_0^1 (N^{x+y}T)^{-t(\alpha+\beta)}\, dt \tag{7}$$

At $\alpha = \beta = -R/L$ (where $L = \log T$), this becomes:

$$(1 + \theta(x+y)) \int_0^1 e^{R[2t+\theta(2t-1)(x+y)]}\, dt \tag{8}$$

After applying the Q operator $Q(-\partial/\partial\alpha)Q(-\partial/\partial\beta)$:

$$Q(\theta t(x+y) - \theta y + t) \times Q(\theta t(x+y) - \theta x + t) \times e^{R[\cdots]} \tag{9}$$

4 The ω Classification: Cases A, B, C

Definition 1 (ω Index)**.** *For piece ℓ with $d = 1$ (single derivative order):*

$$\omega(\ell) := \ell - 2 \tag{10}$$

This gives the classification:

Piece ℓ	$\omega(\ell)$	Case	Structure
1	-1	A	Derivative terms (from d/dx)
2	0	B	No attenuation (direct evaluation)
3	$+1$	C	Auxiliary integral ($\int_0^1 a^{\omega-1} da$)

5 The Six Pair Contributions

The main term c comes from summing over all pairs (ℓ_1, ℓ_2) with $1 \leq \ell_1 \leq \ell_2 \leq 3$:

$$S_{12} = \sum_{\ell_1 \leq \ell_2} (2 - \delta_{\ell_1,\ell_2}) \cdot S_{\ell_1,\ell_2} \tag{11}$$

5.1 Pair (1,1): Case A×A

Both pieces have $\omega = -1$, requiring derivatives.

$$S_{11} = \int_0^1 \int_0^1 (1-u)^2 \left[\frac{d^2}{dx\,dy} P_1(x+u) P_1(y+u) \Big|_{x=y=0} \right] Q(t)^2 e^{2Rt} \, du \, dt \tag{12}$$

The derivative structure:

$$\frac{d^2}{dx\,dy} P_1(x+u) P_1(y+u) \Big|_{x=y=0} = P_1'(u)^2 \tag{13}$$

Contribution: $S_{11} = 0.4129$ (41.5% of baseline total)

5.2 Pair (1,2): Case A×B

Piece 1 has $\omega = -1$ (derivative), piece 2 has $\omega = 0$ (direct).

$$S_{12} = \int_0^1 \int_0^1 (1-u) \left[\frac{d}{dx} P_1(x+u) P_2(u) \Big|_{x=0} \right] Q(t)^2 e^{2Rt} \, du \, dt \tag{14}$$

The mixed structure:

$$\frac{d}{dx} P_1(x+u) \Big|_{x=0} \cdot P_2(u) = P_1'(u) \cdot P_2(u) \tag{15}$$

Contribution:

- Baseline: $S_{12} = +0.4136$ (41.5%)
- Optimized: $S_{12} = +0.3997$ (50.2%) — reduced by cross-term interference

5.3 Pair (1,3): Case A×C

Piece 1 has $\omega = -1$ (derivative), piece 3 has $\omega = +1$ (auxiliary integral).

$$S_{13} = \int_0^1 \int_0^1 \int_0^1 (1-u) P_1'(u) P_3((1-a)u) \cdot a^0 \cdot Q(\cdots)^2 e^{\cdots} \, da \, du \, dt \tag{16}$$

The Case C auxiliary integral arises from (PRZZ Line 2347):

$$\int_{1/q}^1 t^{\alpha+s-1} \log^\tau t \, dt = \frac{(-1)^\tau \tau!}{(\alpha+s)^{\tau+1}} - \cdots \tag{17}$$

Contribution:

- Baseline: $S_{13} = +0.0128$ (+1.3%)
- **Optimized:** $S_{13} = -0.0991$ **(−12.5%) — SIGN FLIP!**

5.4 Pair (2,2): Case B×B

Both pieces have $\omega = 0$, no derivatives or auxiliary integrals.

$$S_{22} = \int_0^1 \int_0^1 P_2(u)^2 \cdot Q(t)^2 e^{2Rt} \, du \, dt \tag{18}$$

This is the simplest case — just polynomial products.
Contribution: $S_{22} \approx 0.140$ (14.0%)

5.5 Pair (2,3): Case B×C

Piece 2 has $\omega = 0$ (direct), piece 3 has $\omega = +1$ (auxiliary integral).

$$S_{23} = \int_0^1 \int_0^1 \int_0^1 P_2(u) P_3((1-a)u) \cdot a^0 \cdot Q(\cdots)^2 e^{\cdots} \, da \, du \, dt \tag{19}$$

Contribution:

- Baseline: $S_{23} = +0.0156$ (+1.6%)
- **Optimized:** $S_{23} = -0.0667$ **(−8.4%) — SIGN FLIP!**

5.6 Pair (3,3): Case C×C

Both pieces have $\omega = +1$, requiring two auxiliary integrals.

$$S_{33} = \int_0^1 \int_0^1 \int_0^1 \int_0^1 P_3((1-a)u) P_3((1-b)u) \cdot a^0 b^0 \cdot Q(\cdots)^2 e^{\cdots} \, da \, db \, du \, dt \tag{20}$$

This is a 4-dimensional integral — the most complex case.
Contribution: $S_{33} \approx 0.001$–0.008 (0.1%–1.0%)

Table 1: Complete pair breakdown for $K = 3$

Pair	ω_1	ω_2	Case	Baseline	Optimized	Δ	Effect
(1,1)	-1	-1	A×A	+0.4129	+0.4129	0.000	—
(1,2)	-1	0	A×B	+0.4136	+0.3997	−0.014	↓
(1,3)	-1	+1	A×C	+0.0128	**−0.0991**	−0.112	↓↓
(2,2)	0	0	B×B	+0.1398	+0.1413	+0.002	↑
(2,3)	0	+1	B×C	+0.0156	**−0.0667**	−0.082	↓↓
(3,3)	+1	+1	C×C	+0.0008	+0.0080	+0.007	↑
	Total S_{12}			0.9954	0.7961	−0.199	**−20%**

6 Summary: The Six Pairs

7 Mirror Term Assembly

Proposition 2 (PRZZ Line 1502)**.**

$$I_1(\alpha,\beta) = I_{1,1}(\alpha,\beta) + T^{-\alpha-\beta} I_{1,1}(-\beta,-\alpha) + O(T/L) \tag{21}$$

At $\alpha = \beta = -R/L$, the factor $T^{-\alpha-\beta} = T^{2R/L} \approx e^{2R/\theta}$.

7.1 The Complete c Formula

The main-term constant combines:

$$\boxed{c = S_{12}(+R) + m \cdot S_{12}(-R) + S_{34}(+R)} \tag{22}$$

where:

- $S_{12}(+R)$ = forward contribution from I_1, I_2
- $S_{12}(-R)$ = mirror contribution (with $R \to -R$)
- $S_{34}(+R)$ = contribution from I_3, I_4 (no mirror needed)
- $m = (f_{I_1} g_{I_1} + (1 - f_{I_1}) g_{I_2}) \cdot (e^R + 2K - 1)$

7.2 Correction Factors

$$g_{I_1} = 1 + \frac{\theta(1-\theta)(2(K-1)+\theta)}{8K(2K+1)^2} = 1.000952 \tag{23}$$

$$g_{I_2} = 1 + \frac{\theta(2-\theta)}{2K(2K+1)} = 1.019436 \tag{24}$$

For $K = 3$, $\theta = 4/7$, $R = 1.3036$:

$$m = 8.814 \tag{25}$$

8 Why Negative Pair Contributions Are Valid

Theorem 1. *Individual pair contributions S_{ℓ_1,ℓ_2} are NOT constrained to be positive.*

Proof sketch.

1. The PRZZ integral structure involves products $P_{\ell_1}(u) \times P_{\ell_2}(u')$
2. For $\ell_1 \neq \ell_2$, the polynomial product can be negative on parts of $[0, 1]$
3. The Case C auxiliary integral introduces additional sign factors via $(-1)^\omega/(\omega - 1)!$
4. Only the **total** $c = \exp(R(1 - \kappa))$ must be positive

□

Physical interpretation: The pairs (1,3) and (2,3) represent cross-terms between different mollifier pieces. These can interfere *destructively*, reducing c and thereby improving κ.

9 Numerical Verification

9.1 Quadrature Convergence

n_{quad}	κ (baseline)	κ (optimized)
40	0.41729593	0.44933435
60	0.41729593	0.44933435
80	0.41729593	0.44933435
100	0.41729593	0.44933435

Table 2: Results stable to $< 10^{-8}$ variation

9.2 R-Sweep Validation

R	κ (baseline)	κ (optimized)	Improvement
1.10	0.338	0.378	+11.9%
1.20	0.382	0.418	+9.4%
1.3036	0.417	0.449	+7.7%
1.35	0.430	0.461	+7.1%
1.40	0.443	0.472	+6.5%

Table 3: Improvement persists across R values (not overfitting)

10 Conclusion

The $\alpha = 61$ polynomial perturbation achieves:

- κ: $0.4173 \to 0.4493$ (**+7.68%**)
- c: $2.137 \to 2.050$ (**−4.09%**)

- S_{12}: $0.995 \to 0.796$ ($-$**20.0%**)

The improvement comes from **destructive interference** in cross-pairs (1,3) and (2,3), which flip from positive to negative contributions. This is mathematically valid within the PRZZ framework.